

Errata for Chapter 1-6 of *Linear Algebra Done Wrong*

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Chapter 1

1. **Page 6, Definition 2.1, last paragraph:** $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_m\mathbf{v}_n = \mathbf{v}$ should be $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \dots + x_n\mathbf{v}_n = \mathbf{v}$.
2. **Page 6, Example 2.2, last displayed formula:** $\mathbf{v} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \dots x_n\mathbf{e}_n = \sum_{k=1}^n x_k\mathbf{e}_k$ should be $\mathbf{v} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \dots + x_n\mathbf{e}_n = \sum_{k=1}^n x_k\mathbf{e}_k$.
3. **Page 8, Definition of linear combination, first paragraph:** $\alpha_k = \mathbf{0} \ \forall k$ should be $\alpha_k = 0 \ \forall k$.
4. **Page 12, “Reflection” example:** The note “see the fig” should be included in the preceding example.
5. **Page 20, First displayed formula:** $\forall x \in \mathbb{F}^r$ should be $\forall \mathbf{x} \in \mathbb{F}^r$.
6. **Page 27, Last line:** $\sum_{k=0}^n a_k t^k, \alpha_k \in \mathbb{F}$ should be $\sum_{k=0}^n a_k t^k, a_k \in \mathbb{F}$.
7. **Page 28, Item 3:** “is isomorphic to $\mathbb{R}^6$ ” should be “is isomorphic to $\mathbb{F}^6$ ”.
8. **Page 36, Footnote 3:** $\mathbb{R}^2$ should be $\mathbb{R}^3$, and 0 should be $\mathbf{0}$.

Chapter 2

1. **Page 54, Paragraph after Proposition 5.1, fourth line:** $n = \dim E$ should be $n = \dim V$.
2. **Page 66, Last paragraph:** $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r, \mathbf{v}_k \in \mathbb{F}^n$ should be $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r \in \mathbb{F}^n$ (first line), and $\mathbb{R}^n$ should be $\mathbb{F}^n$ (last line).

Chapter 3

1. **Page 77, First displayed formula:** “the distance from v_j to $\text{span}\{\mathbf{v}_r : r \neq j\}$ ” should be “the distance from $\mathbf{v}_j$ to $\text{span}\{\mathbf{v}_r : r \neq j\}$ ”.
2. **Page 79, First displayed formula:** Remove the extra equals sign in the first line.
3. **Pages 82-83, Matrix notation:** $A = \{a_{j,k}\}_{j,j=1}^n$ should be $A = \{a_{j,k}\}_{j,k=1}^n$.
4. **Page 88, Second displayed formula and last line of following paragraph:** $D(\mathbf{e}_{j_1}, \mathbf{e}_{j_2}, \dots, \mathbf{e}_{j_n})$ should be $D(\mathbf{e}_{j_1}, \mathbf{e}_{j_2}, \dots, \mathbf{e}_{j_n})$.

5. **Page 91, Last line:** “interchanging rows 2 and 3” should be “interchanging columns 2 and 3”.
6. **Page 94, Second line:** “multiplying j th row of A by j th column of A ” should be “multiplying j th row of A by j th column of C^T ”.
7. **Page 97, Proof of Corollary 6.2, second paragraph:** All instances of k should be r .

Chapter 4

1. **Page 99, Footnote:** $e^{tA} := \sum_{k=0}^{\infty} t^k A^k / k!$ should be $e^{tA} := \sum_{k=0}^{\infty} t^k A^k / k!$.
2. **Page 106, Penultimate paragraph:** $D = S^{-1}AS = [I]_{\mathcal{B}\mathcal{S}}A[I]_{\mathcal{S}\mathcal{B}}$ should be $D = S^{-1}AS = [I]_{\mathcal{B}\mathcal{S}}A[I]_{\mathcal{S}\mathcal{B}}$.
3. **Page 113, Subsection 2.7.2:** $\text{Ker}(A - \lambda T)$ should be $\text{Ker}(A - \lambda I)$.

Chapter 5

1. **Page 127, First line:** $\|\mathbf{v}_k\| \neq 0$ should be $\|\mathbf{v}_k\| \neq 0$.
2. **Page 131, Second line:** $k = 1, 2, \dots, n$ should be $k = 1, 2, \dots, r$.
3. **Page 137, Above second displayed formula:** $A\mathbf{x} \perp \text{Ran } A$ should be $\mathbf{b} - A\mathbf{x} \perp \text{Ran } A$.
4. **Page 144, Proof of Theorem 5.1, penultimate line:** $x \in (\text{Ran } A)^\perp$ should be $\mathbf{x} \in (\text{Ran } A)^\perp$.
5. **Page 153, Proof of Theorem 7.1, last displayed formula:** $T(\mathbf{x} - \alpha\mathbf{y}) = T(\mathbf{x}) + \alpha T(\mathbf{y})$ should be $T(\mathbf{x} + \alpha\mathbf{y}) = T(\mathbf{x}) + \alpha T(\mathbf{y})$.

Chapter 6

1. **Page 165, Theorem 1.2, second line from bottom:** $T = UTU^* = UTU^T$ should be $A = UTU^* = UTU^T$.
2. **Page 167, Proof of Proposition 2.3, first line:** “Theorem 1.1” should be “Theorem 2.1”.
3. **Page 171, Exercise 2.10:** “where α is not a multiple of π ” should be “where $\alpha \neq n\pi$, $n \in \mathbb{Z}$ ”.
4. **Page 173, Section 3.2 title:** “Singular values” should be “Polar decomposition”.
5. **Page 173, Proof of Theorem 3.5, fifth line:** U should be U_0 .
6. **Page 174, Third paragraph, fourth line:** “(defining the *Schur* and singular value decompositions)” should be “(defining the *Schmidt* and singular value decompositions)”.
7. **Page 177, Definition 3.9, third line:** $W \in M_{n \times n}^{\mathbb{F}}, V \in M_{m \times m}^{\mathbb{F}}$ should be $W \in M_{m \times m}^{\mathbb{F}}, V \in M_{n \times n}^{\mathbb{F}}$.

8. **Page 177, Definition 3.9, second paragraph, third line:** $W \in M_{n \times r}^{\mathbb{F}}, V \in M_{m \times r}^{\mathbb{F}}$ should be $W \in M_{m \times r}^{\mathbb{F}}, V \in M_{r \times r}^{\mathbb{F}}$.
9. **Page 177, Note 3.11, third line:** $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ should be $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$.
10. **Page 179, Section 4, first paragraph, second line from bottom:** “(3.5)” should be “(3.4)”.
11. **Page 180, Penultimate paragraph, third line:** σ_1, σ_2 and σ_2 should be σ_1, σ_2 and σ_3 .
12. **Page 180, Third line from bottom:** “(3.1)” should be “(3.4)”.
13. **Page 180, Last displayed formula:** $[A]_{\mathcal{B}\mathcal{A}} = \text{diag}\{\sigma_n : n = 1, 2, \dots, n\}$ should be $[A]_{\mathcal{B}\mathcal{A}} = \text{diag}\{\sigma_k : k = 1, 2, \dots, n\}$.
14. **Page 182, Second paragraph, second line:** “(3.5)” should be “(3.4)”, and $A = W\Sigma V$ should be $A = W\Sigma V^*$.
15. **Page 183, Second line:** “eigenvalues” should be “singular value”.
16. **Page 185, Last displayed formula (after first equality sign):** $\widetilde{W}\widetilde{\Sigma}\widetilde{V}\widetilde{V}^*\widetilde{\Sigma}^{-1}\widetilde{W}^*\mathbf{b}$ should be $\widetilde{W}\widetilde{\Sigma}\widetilde{V}^*\widetilde{V}\widetilde{\Sigma}^{-1}\widetilde{W}^*\mathbf{b}$.
17. **Page 186, First displayed formula:** $y \in \text{Ker } A$ should be $\mathbf{y} \in \text{Ker } A$.
18. **Page 193, Proof of Theorem 5.3, first paragraph:** “Lemma 5.5” should be “Lemma 5.6”, and $U_1 = R_N \dots R_2 R_2 U$ should be $U_1 = R_N \dots R_2 R_1 U$.
19. **Page 194, Fourth paragraph, third line:** $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_n$ should be $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.
20. **Page 197, Exercise 6.4, first line:** $A = A^* > \mathbf{0}$ should be $A = A^* > 0$.